USIU-AFRICA — DSA 3900

Department of Data Science & Analytics

WEEK 6: PROJECT SCOPE AND GANTT CHART

Predicting Maternal Health Risk Among Pregnant Women in Kenya Using Machine Learning and Bayesian Inference on the KDHS 2022

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1. PROJECT SCOPE AND BOUNDARIES

1.1 Scope Definition

This project develops a machine learning-based maternal health risk prediction system for pregnant women in Kenya, drawing exclusively on the Kenya Demographic and Health Survey 2022 (KDHS 2022) Individual Recode dataset. The system classifies each woman as Low, Moderate, or High risk using 13 clinically grounded variables including maternal age, parity, antenatal care visits, place of delivery, wealth index, and pregnancy complications. A Bayesian inference module augments each prediction with a probabilistic confidence score and credible interval. SHAP (SHapley Additive exPlanations) values provide model-level and individual-level explainability. The final deliverable is a Streamlit web application powered by Claude AI (Anthropic API), which translates model outputs into plain-language explanations, care recommendations, and auto-generated referral letters in both English and Swahili, targeting Community Health Workers (CHWs) across Kenya's 47 counties.

1.2 Scope and Exclusion Table

INCLUDED IN SCOPE	EXCLUDED FROM SCOPE
KDHS 2022 Individual Recode (IR) dataset	Primary data collection or patient recruitment
13 maternal health variables selected from KDHS	Native mobile app (iOS/Android) development
Feature engineering — composite risk variable	Real-time data streaming or live EHR integration
ML models: LR, DT, Random Forest, XGBoost, Stacking Ensemble	Integration with NDMA/MOH government systems
Bayesian inference for uncertainty quantification	Clinical trials or validation with real patients
SHAP explainability for model interpretability	Extension to countries outside Kenya
Claude AI for risk explanations and referral letters	Analysis beyond the KDHS 2022 survey period
Streamlit web app — CHW-facing risk scoring tool	Commercial deployment or monetization
County-level risk dashboard (47 counties)	Real-time model retraining pipelines
English and Swahili language support	Electronic Health Record (EHR) system integration

1.3 Key Constraints

- Data is limited to the KDHS 2022 survey period — no real-time data collection is within scope.
- Analysis is restricted to variables available in the KDHS Individual Recode (IR) file.
- No personally identifiable information (PII) is used — all records are fully anonymised by the DHS Program.
- The Streamlit app is a proof-of-concept prototype; full government system integration is excluded.
- The project must be completed within two academic semesters: one proposal, one implementation.

2. KEY TASKS AND MILESTONES

Phase	Milestone	Target Week	Status
Proposal	Topic defined, research gap confirmed	Week 1–2	**Completed**
Proposal	5 research articles summarized	Week 2–3	**Completed**
Proposal	KDHS 2022 dataset accessed & explored	Week 2–4	**Completed**
Proposal	Methodology section drafted	Week 4	**Completed**
Proposal	Expected results & contributions drafted	Week 5	**Completed**
Proposal	Project scope & Gantt chart submitted	Week 6	**In Progress**
Proposal	Concept note presentation delivered	Week 7–8	**Pending**
Proposal	Full proposal document compiled	Week 9–10	**Pending**
Proposal	IRSEC ethical approval submitted	Week 10	**Pending**
Proposal	Proposal defense presented	Week 12	**Pending**
Implement ation	Data preprocessing & feature engineering complete	Sem 2 Wk 4	**Pending**
Implement ation	Baseline models trained & evaluated	Sem 2 Wk 6	**Pending**
Implement ation	Best model selected (stacking ensemble)	Sem 2 Wk 9	**Pending**
Implement ation	Claude AI integration complete	Sem 2 Wk 11	**Pending**
Implement ation	Streamlit app deployed & tested	Sem 2 Wk 13	**Pending**
Implement ation	Final report submitted & defended	Sem 2 Wk 16	**Pending**

3. SCOPE AND TIMELINE NARRATIVE

The project is structured across two academic semesters: the current Proposal Phase (Summer Semester 2026, Weeks 1–12) and the subsequent Implementation Phase (Fall Semester 2026, Weeks 1–16). This two-phase structure ensures that the research foundation is rigorously established before any technical development begins, in alignment with USIU-Africa's DSA 3900 course requirements and standard data science research practice.

During the Proposal Phase, the primary focus is on defining the research problem, establishing the literature base, designing the methodology, and obtaining ethical clearance through IRSEC. Weeks 1 through 5 have been dedicated to topic refinement, literature review, dataset exploration on the KDHS 2022 portal, and drafting the methodology and expected contributions sections. Week 6 marks the midpoint of the proposal phase, with the project scope and Gantt chart formalizing the research boundaries and timeline. Weeks 7 through 12 will cover the concept note presentation, full proposal document compilation, IRSEC submission, and the final proposal defense.

The Implementation Phase begins in Semester 2 and follows a sequential pipeline: data preprocessing and feature engineering (Weeks 1–4), baseline model development (Weeks 4–6), advanced model development and hyperparameter tuning using Optuna (Weeks 6–9), Bayesian inference and SHAP integration (Weeks 9–10), Claude AI and Streamlit app development (Weeks 10–12), testing and validation (Weeks 12–13), and final report writing and defense preparation (Weeks 13–16). Task dependencies are designed so that each phase builds directly on the output of the prior one, ensuring that no implementation step is attempted without the preceding foundation being complete.

This timeline is realistic and achievable within the constraints of the project. The KDHS 2022 dataset is already downloaded and loaded in Google Colab, the variable selection and target variable engineering logic are already defined, and a proof-of-concept Claude AI-powered web application prototype has been developed. These early deliverables de-risk the implementation timeline significantly, as the two most common project failure points — data access delays and end-product design uncertainty — have already been resolved in the proposal phase.

4. PROJECT PLAN OUTLINE

4.1 Research Guidelines

- All analysis will be conducted on the anonymised KDHS 2022 dataset in compliance with the DHS Program data use agreement.
- IRSEC ethical approval will be obtained prior to commencement of the implementation phase.
- All model results and performance metrics will be reported transparently, including failure cases.
- The project will follow a reproducible research workflow — all code will be version-controlled on GitHub.

4.2 Key Milestones Summary

- Proposal phase completed and defended — Week 12, Summer 2026
- Data preprocessing pipeline complete — Week 4, Semester 2
- Final model selected and validated — Week 9, Semester 2
- Claude AI-powered Streamlit app deployed — Week 12, Semester 2
- Final report submitted and project defended — Week 16, Semester 2

4.3 Resources Required

Resource	Description	Availability
KDHS 2022 Dataset	Primary data source — 32,156 records, 5,925 variables	Downloaded
Google Colab	Cloud computing environment for Python analysis	Free — Active
Python (scikit-learn, PyMC, SHAP, XGBoost, CatBoost)	ML, Bayesian, and explainability libraries	Free — Open Source
Streamlit	Web application framework for end product	Free — Open Source
Anthropic Claude API	AI explanations and referral letter generation	API Access Required
Google Drive	Dataset storage and notebook backup	Free — Active
GitHub	Version control and code repository	Free — Active
Supervisor	Academic guidance and milestone review	USIU-Africa Faculty

GANTT CHART — FULL PROJECT TIMELINE

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Active / In Progress | Not Active

PHASE 1 — PROPOSAL (Summer Semester 2026)

Task	W1	W2	W3	W4	W5	W6	W7	W8	W9	W10	W11	W12
Topic Selection & Problem Definition	■	■										
Literature Review		■	■									
Dataset Access & Exploration (KDHS 2022)		■	■	■								
Methodology Section				■								
Expected Results & Contributions					■							
Project Scope & Gantt Chart						■						
Concept Note Preparation							■					
Concept Note Presentation								■				
Full Proposal Document Compilation									■	■		
IRSEC Ethical Approval Submission										■		
Proposal Presentation Preparation											■	
Proposal Defense & Submission												■

PHASE 2 — IMPLEMENTATION (Fall Semester 2026 — Weeks 1–12 shown, continues to Week 16)

Task	W1	W2	W3	W4	W5	W6	W7	W8	W9	W10	W11	W12
Environment Setup & Data Loading	■	■										
Data Preprocessing & Feature Engineering		■	■	■								
Baseline ML Models (LR,				■	■							

Decision Tree)				■	■								
Advanced Models (RF, XGBoost, CatBoost, Ensemble)					■	■	■	■					
Bayesian Inference Integration (PyMC)							■	■	■				
SHAP Explainability									■	■			
Claude AI Integration (Anthropic API)										■	■		
Streamlit App Development & Dashboard											■	■	
Testing, Validation & Peer Review												■	
Final Report Writing												■	

Note: Implementation Phase continues in Weeks 13–16 with Final Report Submission (Wk 14), Presentation Preparation (Wk 15), and Final Defense (Wk 16). Gantt extended to Week 12 for display.